8. Japanese pine sawyer beetle (*Monochamus alternatus*)

Last modified: October 16, 2026

8A. Introduction

The Japanese pine sawyer beetle (JPSB), Monochamus alternatus Hope (Coleoptera: Cerambycidae), is a major vector of nematodes that cause pine wilt disease, particularly Bursaphelenchus xylophilus (Vicente et al. 2012, EPPO 2024). Native to East Asia, including mainland China, Taiwan, Laos, Korea, and Japan, JPSB has expanded its range to other parts of Asia, including Vietnam (Kobayashi et al. 1984, Ma et al. 2006, EPPO 2024). Adult beetles acquire and transmit nematodes while feeding on healthy pine trees, with infection disrupting xylem function and causing wilt and, in susceptible trees, death within weeks (Kobayashi et al. 1984, Vicente et al. 2012). Females primarily oviposit in weakened or diseased trees, where larvae develop beneath the bark and in the sapwood, nematodes subsequently associate with newly emerging adults, completing the transmission cycle (Kobayashi et al. 1984, Ma et al. 2006, An et al. 2019). Although B. xylophilus is native to North America and generally causes little damage in its native range, JPSB could also vector other nematode species not currently present in North America (CAPS 2013, An et al. 2019). JPSB primarily attacks conifers, particularly Pinus, Abies, Picea, Larix, and Cedrus, and pine wilt disease has caused extensive tree mortality and substantial economic losses where it has been introduced, particularly in Asia and Europe (Vicente et al. 2012, An et al. 2019).

Areas suitable for JPSB are predicted throughout much of the southern and midwestern United States, whereas cold temperatures may limit establishment in colder regions, including parts of the Intermountain West and Northeast (Song and Xu 2006, Barker and Coop 2020), Gao et al. 2023). Parts of the West Coast, including most of California, were also predicted to be suitable by some models (Song and Xu 2006, Barker and Coop 2020). Within environmentally suitable areas of the contiguous United States, JPSB is predicted to complete at least one generation annually, with up to four generations in the South (Barker and Coop 2020). In colder parts of its native range, JPSB may be univoltine or semivoltine, whereas 2–3 generations can occur annually in warmer regions such as subtropical and tropical China (Togashi 1989, Ma et al. 2006, CAPS 2013). JPSB undergoes winter diapause primarily during the final larval instar, with diapause induction influenced by larval density (Togashi 2021).

A model for Japanese pine sawyer beetle developed for the Degree-Day, establishment Risk, and Phenological event mapping system (DDRP) is operationalized to produce near real-time forecasts for the contiguous United States (CONUS) at https://uspest.org/DDRP. The phenology model applies a lower developmental threshold of 12.2 °C and upper threshold of 35 °C for all life stages (egg, larva, pupa, and adult) and assumes the larva is the overwintering stage. The climatic suitability model applies a lower cold stress threshold and upper heat stress threshold of −4 °C and 36 °C, respectively. The full set of model parameters for Japanese pine sawyer beetle, including degree-day requirements for individual life stages and climate stress limits, are presented in Table 1. Details on model parameterization are available at USPest.org (Barker and Coop 2020).

Table 1. Japanese pine sawyer beetle model parameters for DDRP.

Parameter	Code	Value
Lower developmental thresholds (°C)		
Egg	`eggLDT`	12.2
Larvae	`larvaeLDT`	12.2
Pupae	`pupaeLDT`	12.2

Parameter	Code	Value
Adult	adultLDT	12.2
Upper developmental thresholds (°C)		
Egg	eggUDT	35.0
Larvae	larvaeUDT	35.0
Pupae	pupaeUDT	35.0
Adult	adultUDT	35.0
Stage durations (°C degree-days)		
Egg	eggDD	83
Larvae	larvaeDD	647
Pupae	pupDD	166
Adult	adultDD	207
Pest events (°C degree-days)		
J-larval development complete	OWEventDD	—
Egg hatch	eggEventDD	80
Developing J-larvae	larvaeEventDD	320
Adult emergence (pupal development complete)	pupaeEventDD	83
Egg laying	adultEventDD	107
Cold stress		
Cold stress temperature threshold (°C)	coldstress_threshold	-4
Cold degree-day (°C) limit when most individuals die	coldstress_units_max1	525
Cold degree-day (°C) limit when all individuals die	coldstress_units_max2	800
Heat stress		
Heat stress temperature threshold (°C)	heatstress_threshold	36
Heat stress degree-day (°C) limit when most individuals die	heatstress_units_max1	800
Heat stress degree-day (°C) limit when all individuals die	heatstress_units_max2	1100
Cohorts		
Degree-days (°C) to emergence (average)	distro_mean	204
Degree-days (°C) to emergence (variation)	distro_var	3000
Minimum degree-days (°C) to emergence	xdist1	10
Maximum degree-days (°C) to emergence	xdist2	410
Shape of the distribution	distro_shape	normal
Other		
Order of stages	stgorder	OL, P, A, E, L
Degree-day calculation method	calctype	triangle

8B. Results

i. Phenology

Entire period: 1981–2025

Over 1981–2025, predicted first adult emergence and egg hatch generally shifted earlier across much of the southern and eastern United States, while later events occurred in portions of the northern interior (Figure 1, Figure 2). Significant changes were concentrated in the eastern and central United States, with additional localized changes in the West.

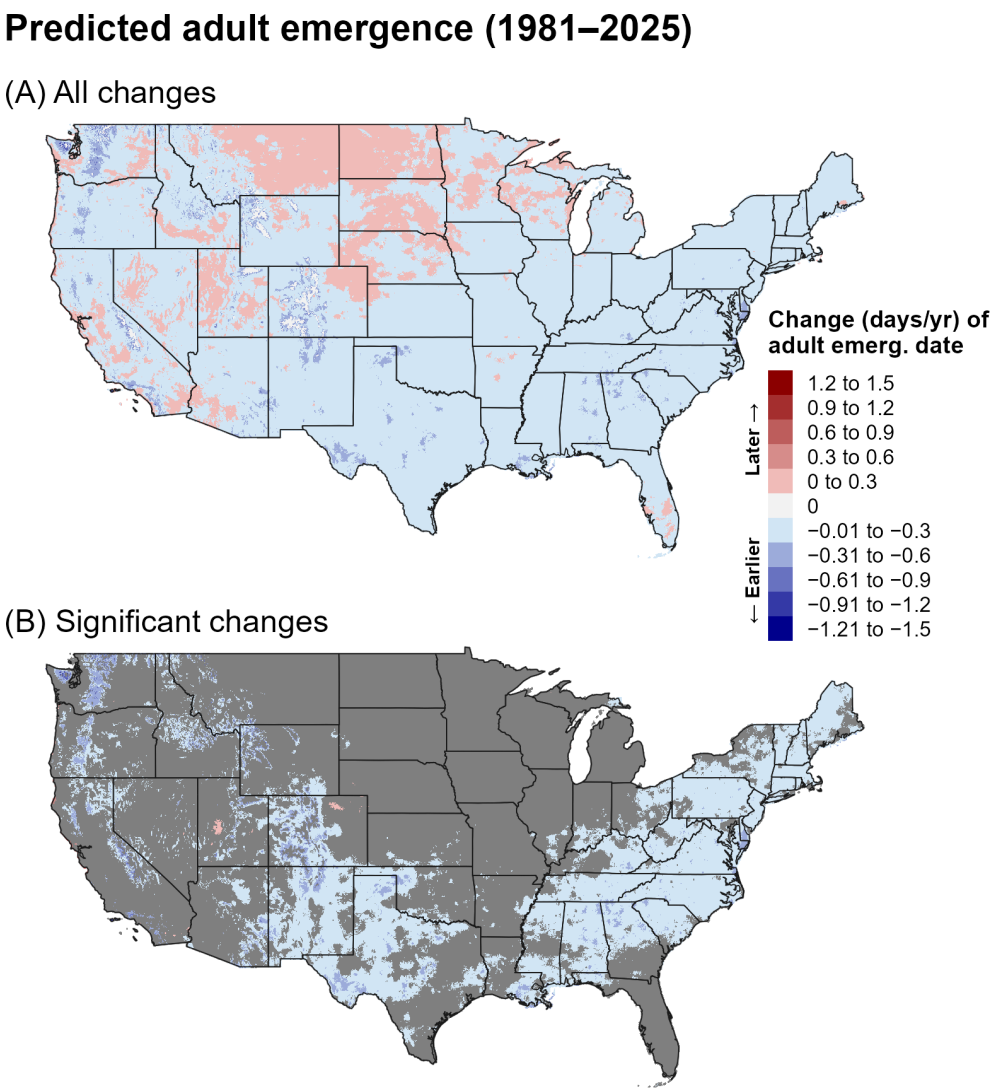


Figure 1. Changes (days/year) in predicted dates of first adult emergence for Japanese pine sawyer beetle across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

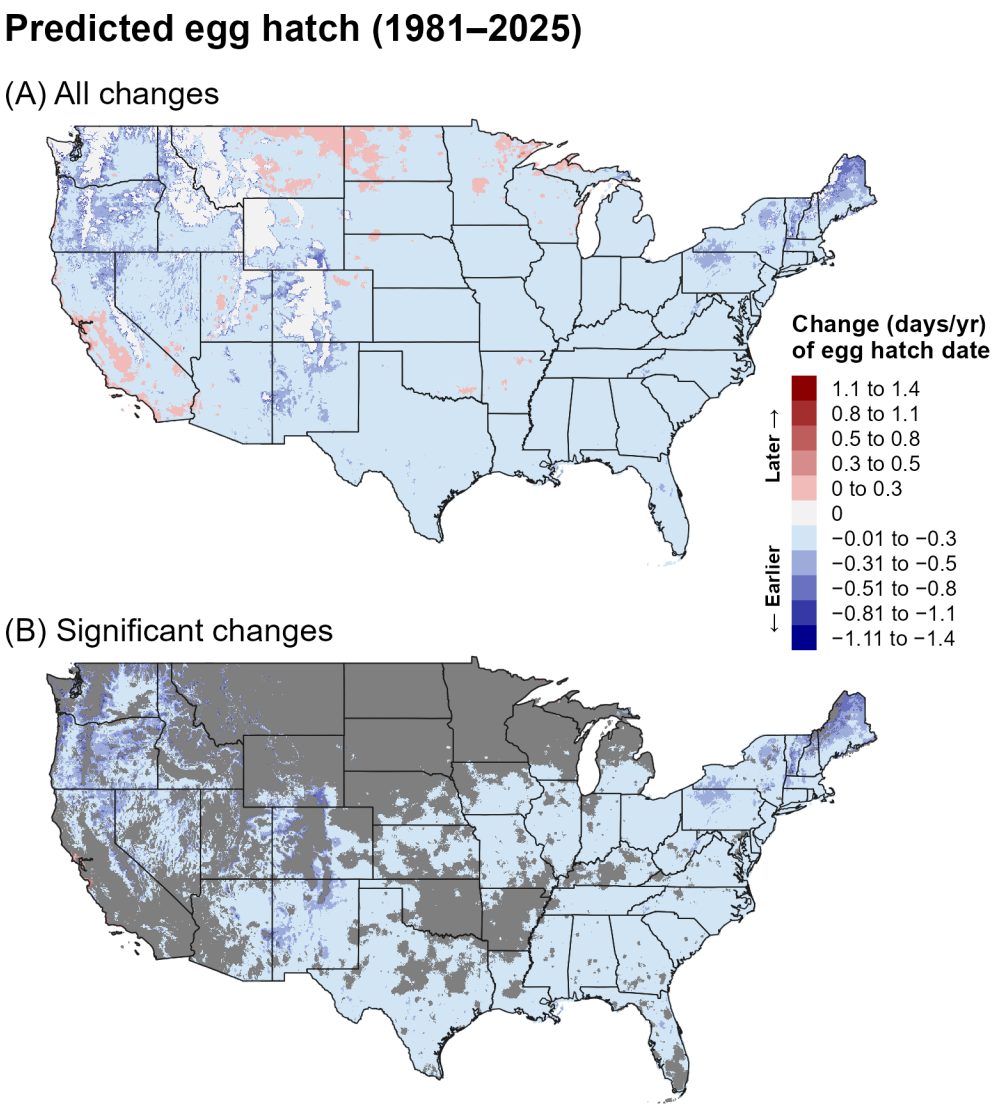


Figure 2. Changes (days/year) in predicted dates of first egg hatch for Japanese pine sawyer beetle across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

Recent period: 2001–2025

During 2001–2025, earlier phenological events remained apparent across much of the western and central United States, while later events occurred in parts of the northern interior and East (Figure 3, Figure 4). Significant changes were spatially limited compared with the full period.

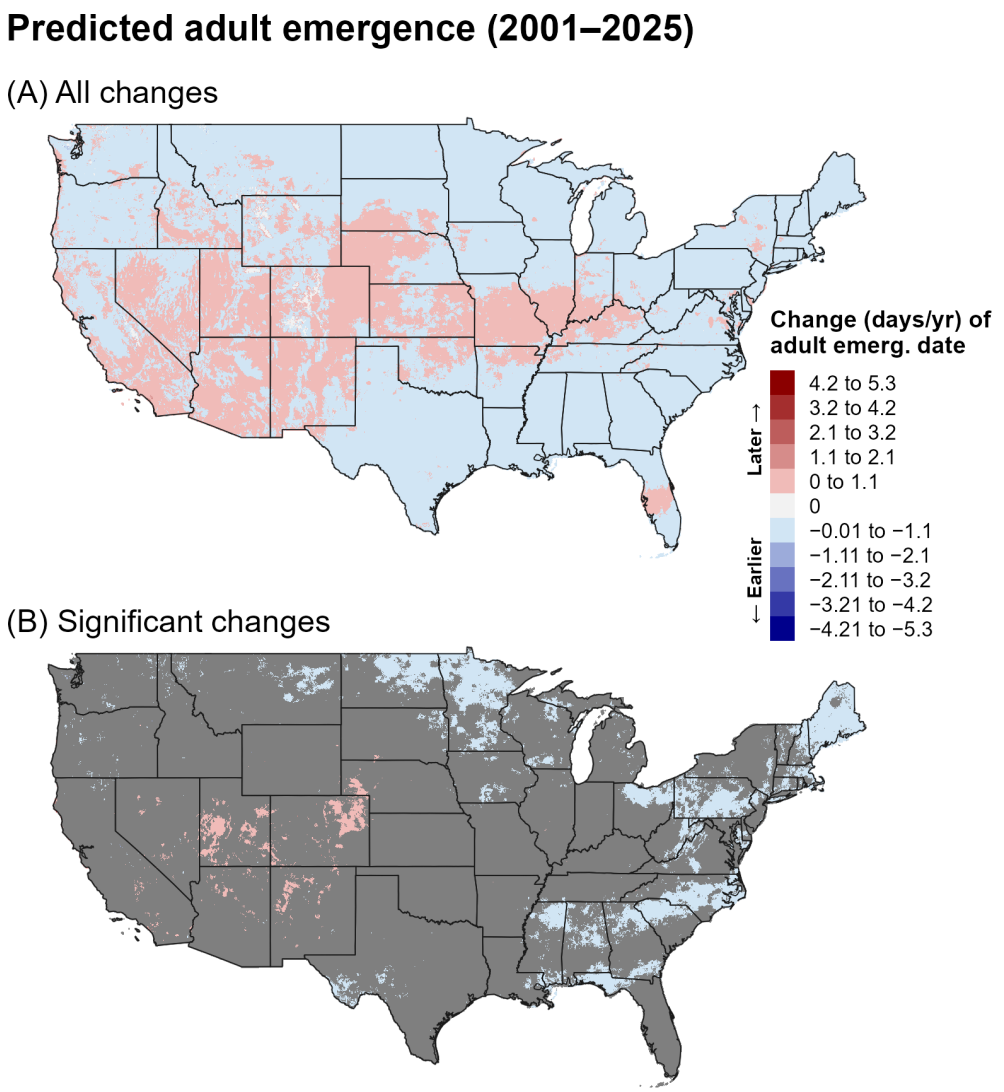


Figure 3. Changes (days/year) in predicted dates of first adult emergence for Japanese pine sawyer beetle across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

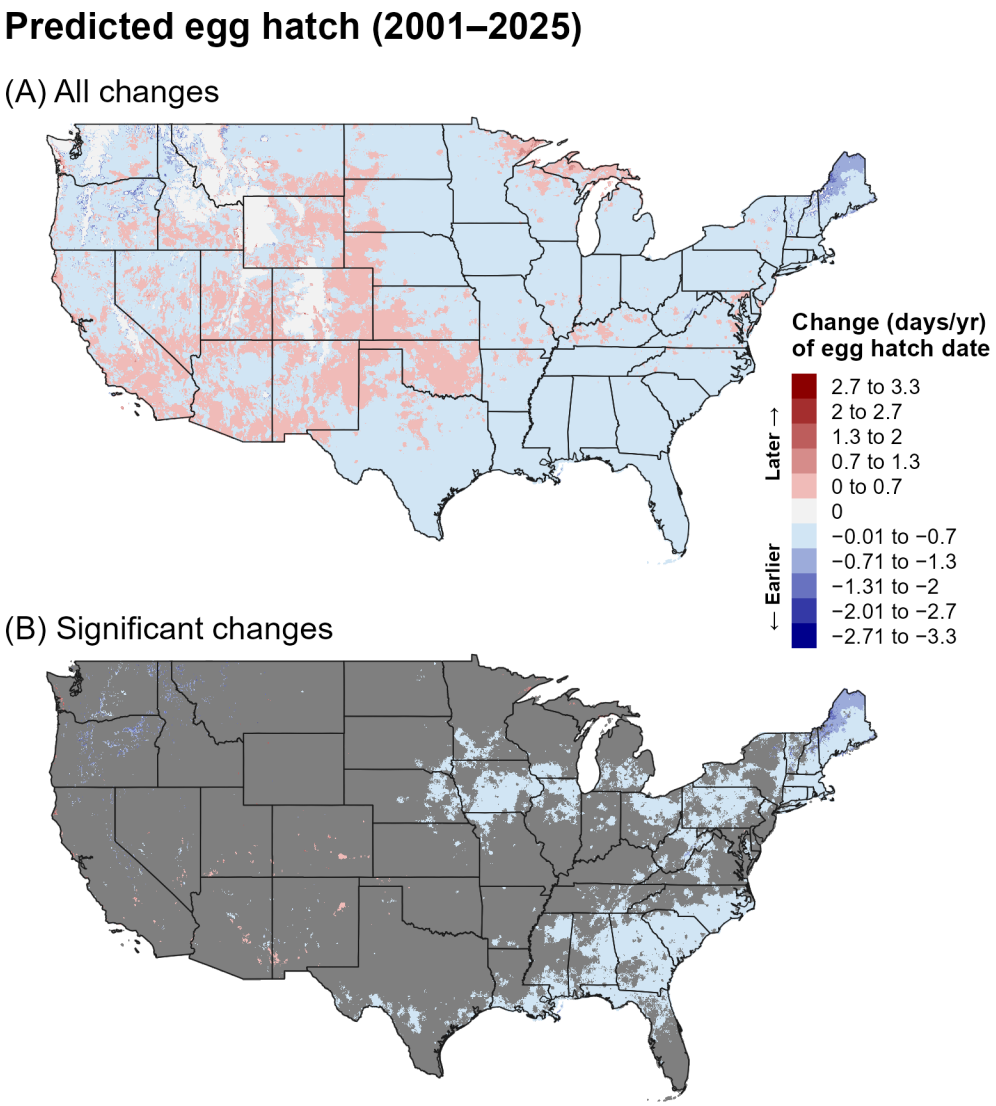


Figure 4. Changes (days/year) in predicted dates of first egg hatch for Japanese pine sawyer beetle across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

ii. Climatic suitability

Entire period: 1981–2025

Predicted cold stress decreased across much of the northern United States over 1981–2025, with significant decreases in portions of the northern Rockies, Great Lakes, and Northeast (Figure 5). Heat stress increased across the Southwest and southern Great Plains (Figure 6). Climatic suitability increased significantly in portions of the northern United States, particularly the northern Plains, Great Lakes, and Northeast (Figure 7).

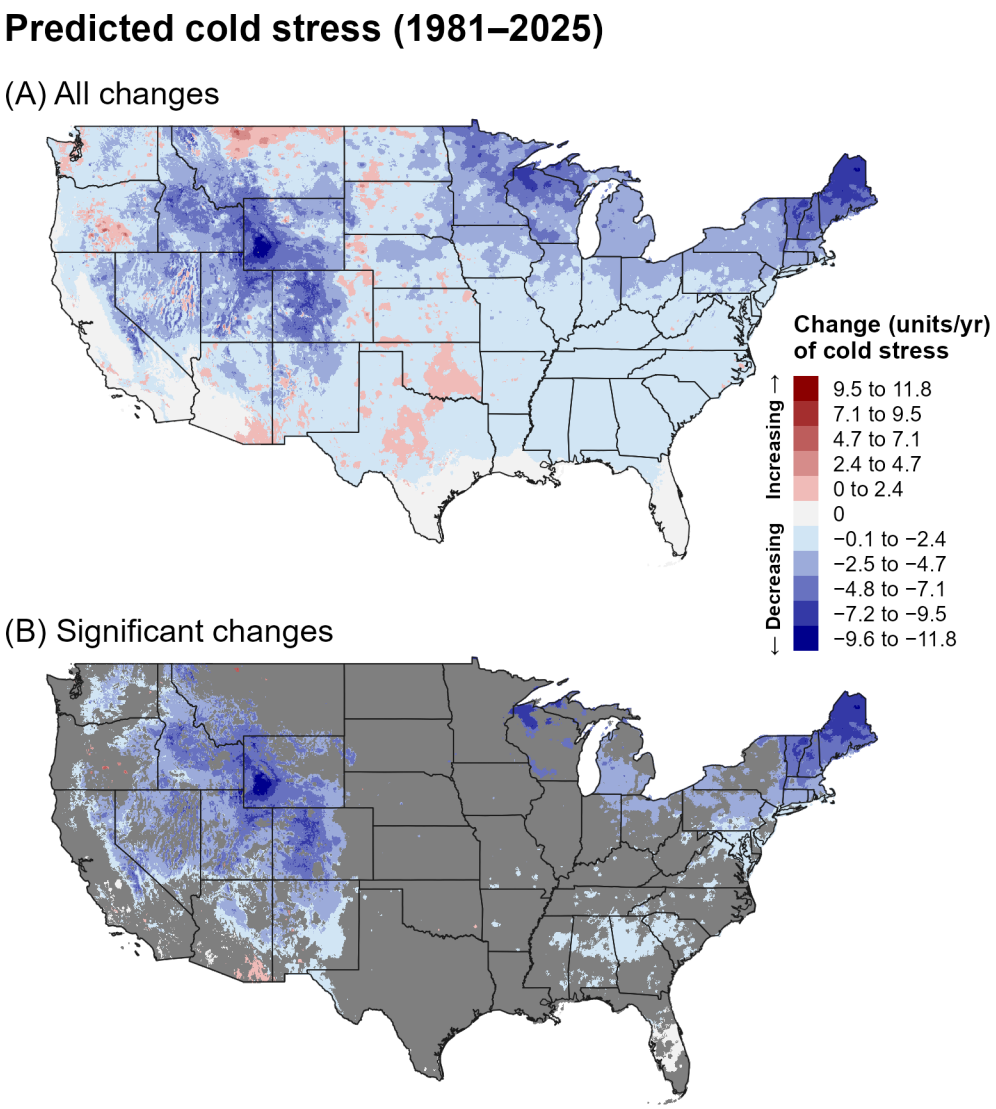


Figure 5. Changes (units/year) in predicted cold stress for Japanese pine sawyer beetle across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

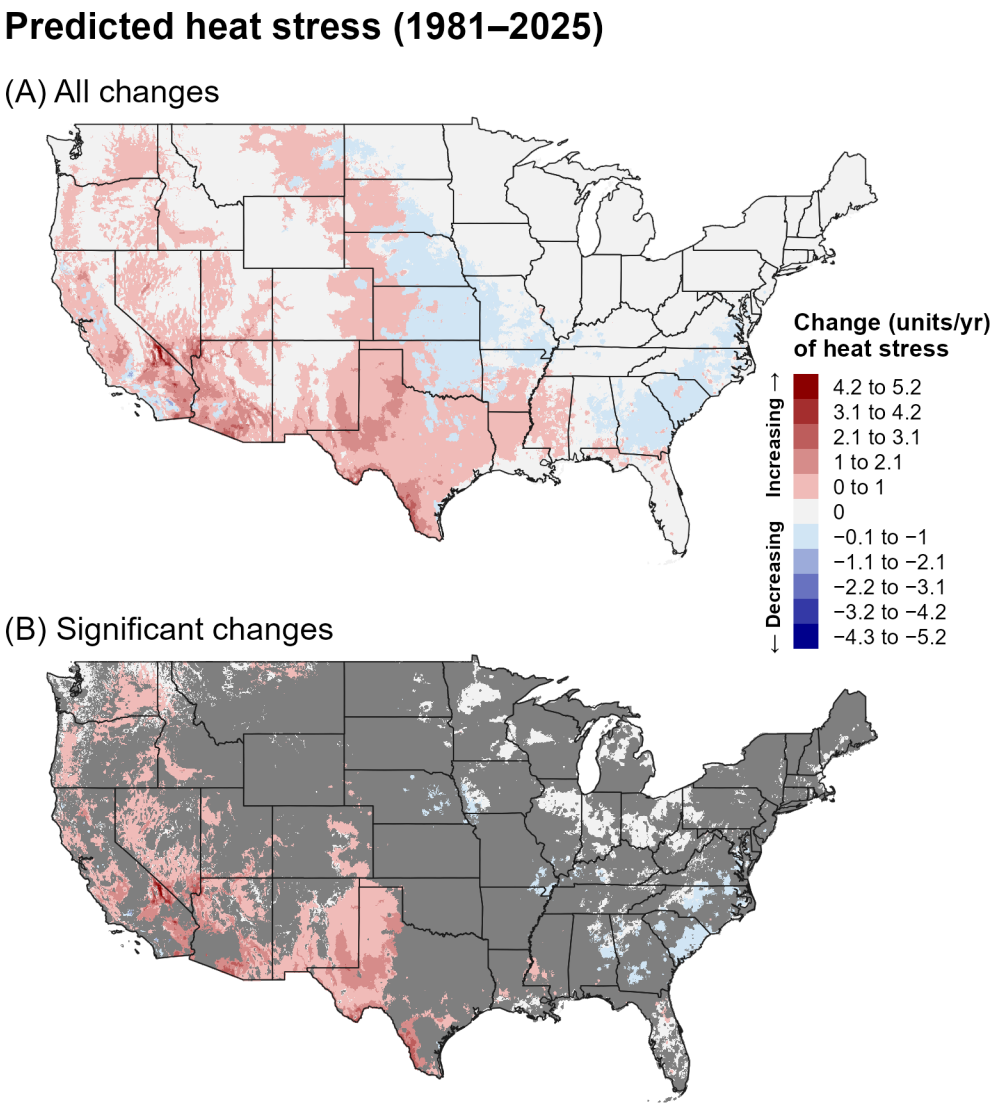


Figure 6. Changes (units/year) in predicted heat stress for Japanese pine sawyer beetle across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

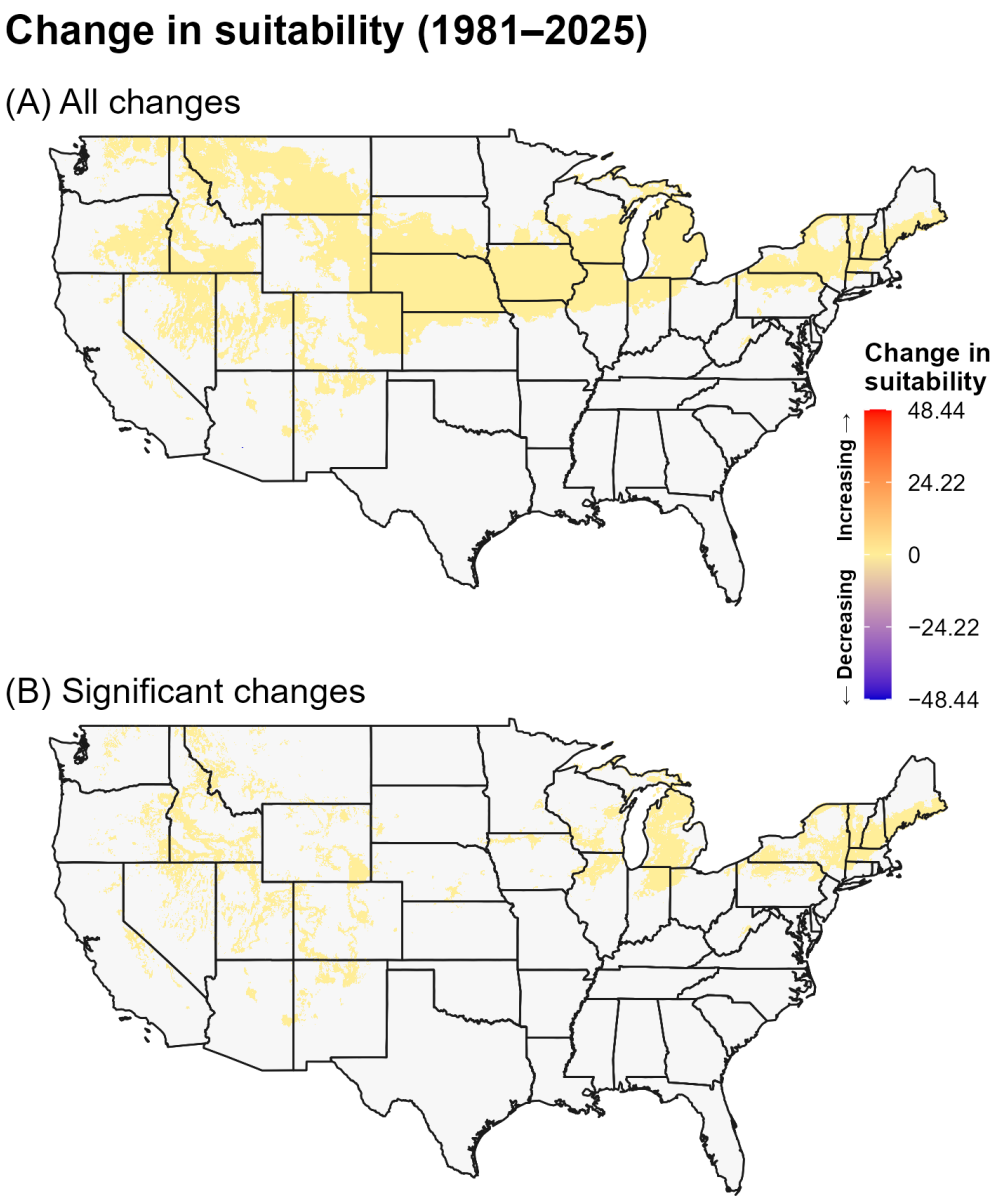


Figure 7. Changes in predicted climatic suitability for Japanese pine sawyer beetle across the CONUS over the 44-year period (1981–2025). Suitability is defined as areas not excluded by climate stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

Recent period: 2001–2025

During 2001–2025, increases in cold stress became more evident across portions of the northern and central interior, although decreases persisted in scattered northern and eastern areas (Figure 8). Heat stress continued to increase in the Southwest and southern Great Plains (Figure 9). Significant suitability increases were more spatially limited during the recent period but remained evident in portions of the northern United States (Figure 10).

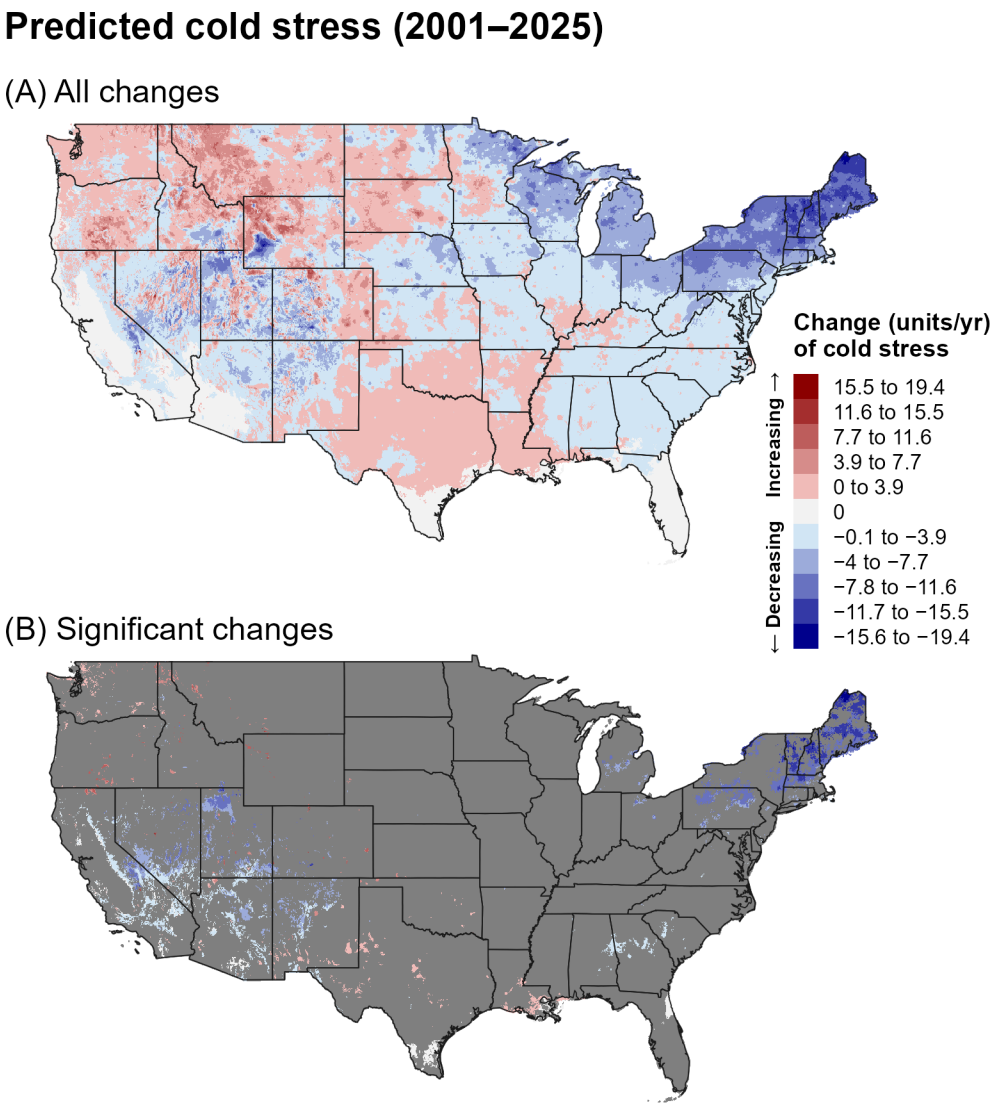


Figure 8. Changes (units/year) in predicted cold stress for Japanese pine sawyer beetle across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

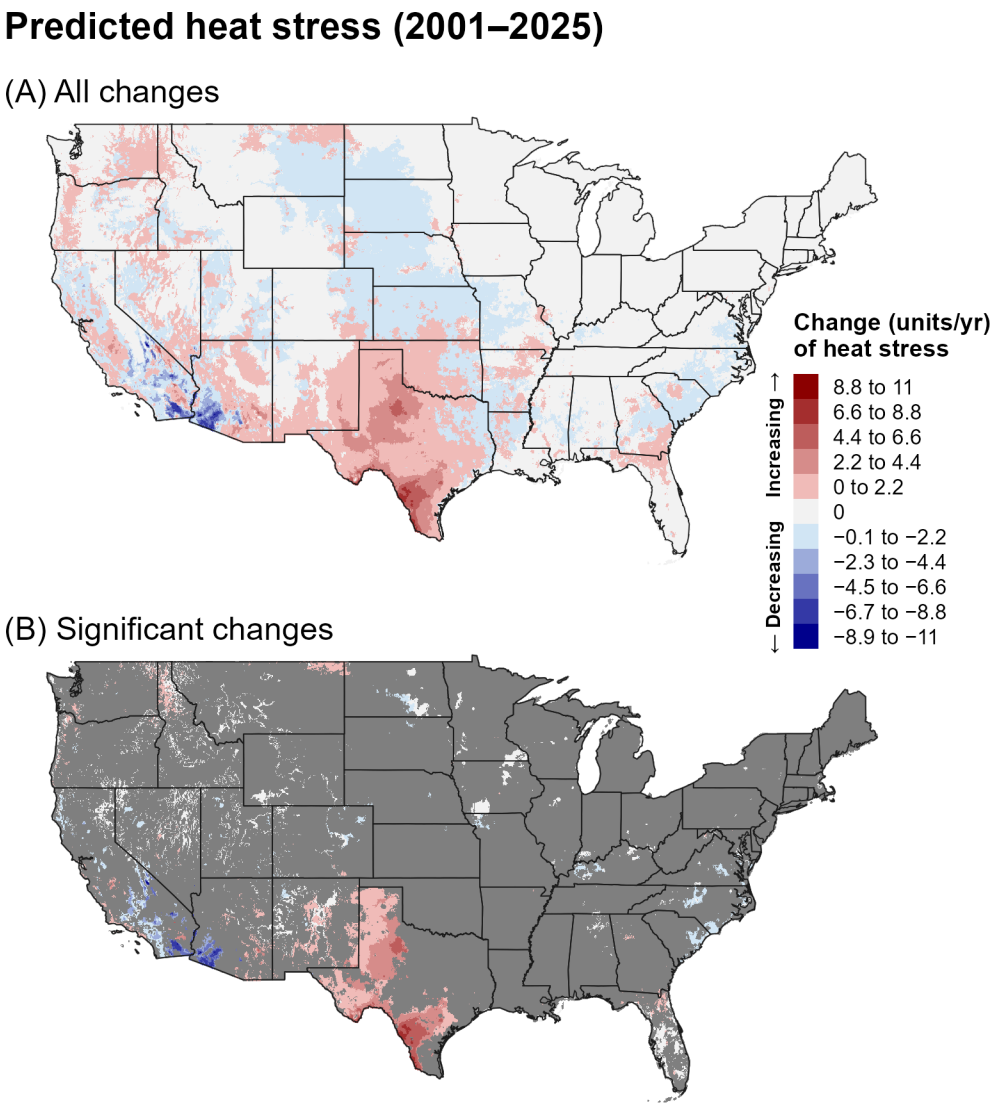


Figure 9. Changes (units/year) in predicted heat stress for Japanese pine sawyer beetle across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

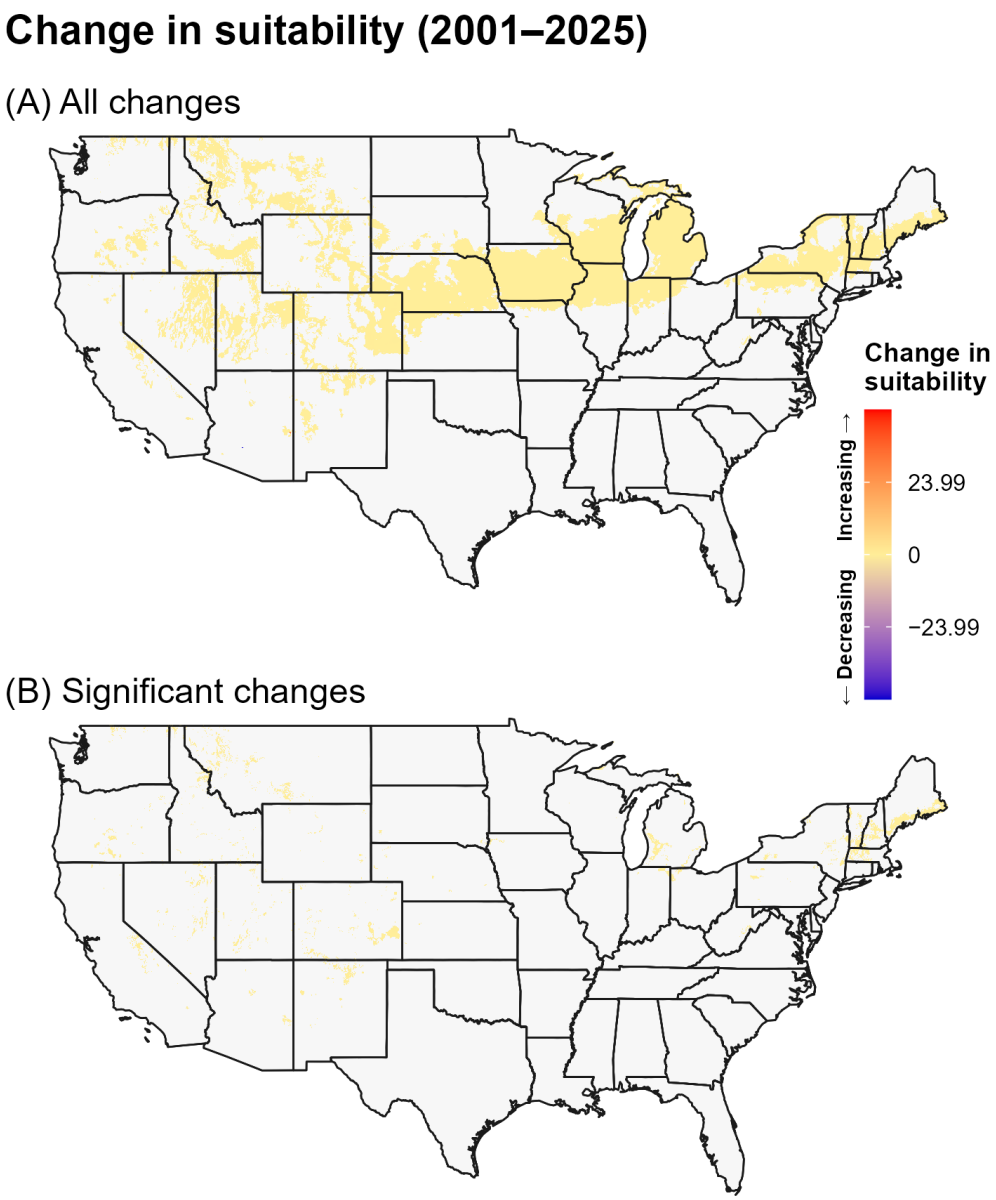


Figure 10. Changes in predicted climatic suitability for Japanese pine sawyer beetle across the CONUS over the recent 24-year period (2001–2025). Suitability is defined as areas not excluded by climate stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

8C. Discussion

The JPSB results indicate earlier phenology across many regions over the full period, with more limited and heterogeneous changes during 2001–2025. Because JPSB is associated with conifer hosts and its development is temperature dependent, regional shifts in emergence and egg hatch could alter the timing of detection and management activities. The stronger long-term pattern suggests that recent variability may be obscuring some of the longer-term directional signal.

The increase in predicted climatic suitability across portions of the northern United States over the full period is consistent with the concurrent decline in cold stress in those regions. This pattern suggests that

warming may reduce cold-related limitations on establishment. During the recent period, increases in cold stress in parts of the northern interior corresponded with weaker suitability changes, illustrating how shorter-term regional climate patterns can alter the establishment signal.

The stronger and more spatially extensive trends in predicted phenology and climate stress over the full 44-year period may reflect both the larger sample size, which can reduce noise and improve detection of directional patterns, and differences in the underlying climate trends between the two time periods. In particular, the recent 24-year period may capture climate variability or regional trends that differ from those observed over the longer 44-year period. Thus, differences between the two periods should be interpreted as reflecting both long-term climate trends and shorter-term regional variability.

8D. References

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